

REPORT
Of
DELOITTE DATA ANALYTICS & FORENSIC TECHNOLOGY
JOB SIMULATION

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In partial fulfilment of the requirement of degree of
B.Tech AIML



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CANDIDATE'S DECLARATION

I, Rahul Sharma, do hereby declare that the internship report titled “Deloitte Data Analytics & Forensic Technology Job Simulation” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in AIML. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

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I extend my practical appreciation to Deloitte Australia and Forage for designing an insightful and challenging job simulation program that mirrors real-world consulting and technology audits.

I express my gratitude to my parents for their blessings.

INTERNSHIP COMPLETION CERTIFICATE

The official Certificate of Completion issued by Deloitte Australia and Forage upon successful completion and verification of both analytical and forensic work modules is attached below:

Deloitte.

Rahul Sharma
Data Analytics Job Simulation

Certificate of Completion
August 30th, 2026

Over the period of August 2026, Rahul Sharma has completed practical tasks in:

Data analysis
Forensic technology



Tina McCreery
Chief Human
Resources Officer,
Deloitte

Enrolment Verification Code 6a92d70962825647729c7298 | User Verification Code 6a79575b02955c8d92b6bce8 | Issued by Forage

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CHAPTER 4: PROJECT DESCRIPTION

4.1 Introduction

In the contemporary corporate technology landscape, enterprise analytics and forensic audit procedures play a pivotal role in maintaining operational efficiency and corporate compliance. The Deloitte Data Analytics & Forensic Technology Virtual Job Simulation offers a realistic client consulting environment designed to mirror the complex engagements executed by Deloitte's Risk Advisory and Forensic teams.

This internship project comprised two specialized work modules. The first module focused on high-frequency IoT factory telemetry analytics for global manufacturing client Daikibo, leveraging Tableau to process millions of diagnostic log signals to isolate critical failure nodes. The second module focused on corporate forensic technology and HR audit compliance, utilizing Microsoft Excel conditional formulas to analyze employee compensation equity across global sites and job roles.

4.2 Organization Profile

Deloitte Touche Tohmatsu Limited (Deloitte) is one of the world's leading professional services networks, providing audit, consulting, financial advisory, risk advisory, tax, and forensic technology services across more than 150 countries and territories. With a global workforce exceeding 450,000 professionals, Deloitte serves four out of five Fortune Global 500 companies.

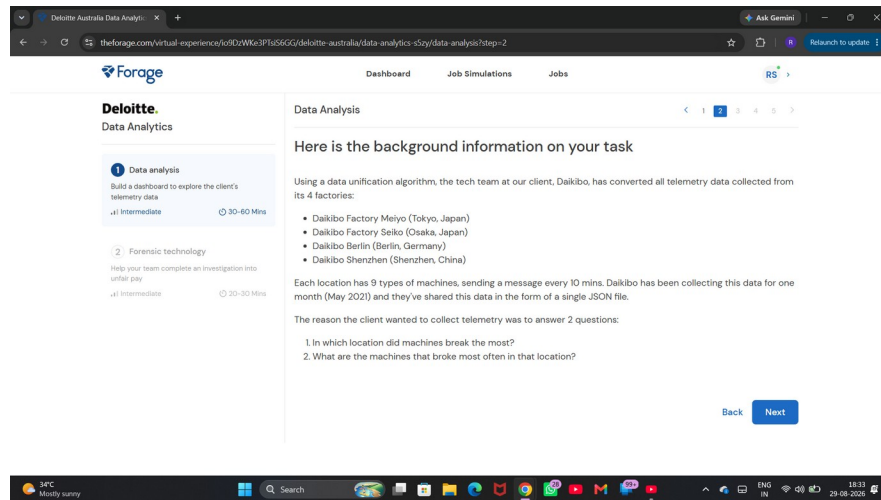
Deloitte Risk Advisory & Forensic Technology Practice delivers specialized technical solutions to help organizations manage financial, operational, and regulatory risks. The division combines visual data analytics, automated reporting, and algorithmic forensic audits to resolve complex corporate challenges, including supply chain bottlenecks and wage parity investigations.

4.3 Problem Statement

The job simulation presented two distinct operational challenges across separate corporate domains:

Problem Statement 1 (Factory Telemetry Analytics): International manufacturing client Daikibo collected continuous operational telemetry across four major production plants: Daikibo Factory Meiyo (Tokyo, Japan), Daikibo Factory Seiko (Osaka, Japan), Daikibo Berlin (Berlin, Germany), and Daikibo Shenzhen (Shenzhen, China). Each location operates 9 distinct heavy machinery types that transmit status updates every 10 minutes (aggregated into a single JSON file for May 2021). Daikibo leadership required definitive analytical answers to two key questions: (1) In which plant location did machines break down most frequently? (2) What specific machine types experienced the highest failure frequency in that location?

Figure 4.1: Deloitte Data Analytics Task 1 Portal Screenshot (Telemetry Analytics)



Problem Statement 2 (Forensic Pay Equity Audit): The forensic technology team was assigned to investigate employee compensation data ('Equality Table.xlsx') comprising Factory Location, Job Role, and Equality Score (integer values between -100 and +100, where 0 represents optimal equity). The objective was to design an automated 4th column ('Equality Class') that categorizes employees into three compliance tiers: Fair (+10 to -10), Unfair (< -10 AND > 10), and Highly Discriminative (< -20 AND > 20).

Figure 4.2: Deloitte Forensic Technology Task 2 Portal Screenshot (Pay Equity Audit)

Deloitte Australia Data Analytics

Tableau Step Guide

Tableau Job Simulation Guide

Ask Gemini

theforage.com/virtual-experience/IOl8ZWe3Pls2SGG/deloitte-australia/data-analytics<5zy/forensic-technology/step=3

Relaunch to update

Deloitte

Data Analytics

✔ Data analysis

Build a dashboard to explore the client's telemetry data

✔ Intermediate

30-60 Mins

🔍 Forensic technology

Help your team complete an investigation into unfair pay

✔ Intermediate

30-30 Mins

Forensic technology

Here is your task

We have processed all data on employee compensation and generated an Excel file (Equality Table.xlsx, available in the Resources) containing 3 columns:

1. Factory
2. Job Role
3. Equality Score (integer; ranging between -100 and +100; 0 is ideal)

Here is your task:

- Create a 4th column (Equality class), classifying the equality score into 3 types:
 - Fair (-<10)
 - Unfair (-<-10 AND +10)
 - Highly Discriminative (-<-20 AND +20)

Examples:

- 10 → Fair
- -9 → Unfair
- -30 → Highly Discriminative

Please find the Equality Table you need to edit in the resources below. When you are done, upload the edited version of the file.

Equality Table

Download File

32°C
Partly cloudy

Search

20:31
29-08-2020

4.4 Project Objectives

The primary technical and business objectives accomplished during this project include:

1. Telemetry Log Ingestion: Parse, aggregate, and transform unstructured JSON telemetry log streams from 4 global manufacturing facilities.
2. Interactive Tableau Dashboarding: Construct dynamic bar charts and comparative matrices in Tableau to isolate peak operational breakdown locations.
3. Forensic Classification Modeling: Formulate robust, error-free nested conditional logic in Microsoft Excel ('IF', 'AND', 'OR') to audit salary equity scores across multi-national employee pools.
4. Actionable Executive Reporting: Synthesize analytical insights into executive-ready presentation slide decks and structured audit reports.

4.5 Scope of the Project

- In-Scope: Processing 31 days of May 2021 machine status logs; evaluating 9 heavy machinery categories across Tokyo, Osaka, Berlin, and Shenzhen; Tableau data visual dashboard creation; nested Excel logic design for equality scores ranging from -100 to +100; online platform submission and verification.
- Out-of-Scope: Physical machine repairs or hardware servicing on factory floors; continuous live stream IoT integration; manual payroll modifications or bank disbursement processing.

4.6 Technologies and Tools Used

- Tableau Desktop / Tableau Public: Used for visual data discovery, interactive dashboard layout, calculated fields creation, and dimensional aggregation of JSON telemetry logs.
- Microsoft Excel: Deployed for forensic financial modeling, nested conditional classification logic ('IF', 'AND', 'OR'), and tabular formatting.
- JSON (JavaScript Object Notation): Telemetry log format representing time-series machine status broadcasts.
- Forage Portal: Enterprise simulation platform used for submission, automated evaluation, and verification.

4.7 System Architecture

The technical architecture follows a dual-track analytical processing workflow, as illustrated in Figure 4.3 below:

Figure 4.3: End-to-End Analytics & Forensic Audit Data Pipeline Architecture

Track 1: Telemetry Data Analytics Pipeline (Tableau)

[4 Global Production Plants (JSON Telemetry Logs)] → [Data Unification & Ingestion Engine] → [Tableau ETL & Aggregation] → [Interactive Visual Dashboard] → [Factory Bottleneck Identification]

Track 2: Forensic Pay Equity Audit Pipeline (Excel)

[Raw Payroll Dataset (`Equality Table.xlsx`)] → [Data Cleaning & Schema Inspection] → [Nested Logic Engine (`IF`/`AND`/`OR`)] → [Equality Class Population] → [Executive Compliance Summary]

4.8 Methodology

4.8.1 Task 1 Methodology: Telemetry Data Analytics

1. Data Import & Cleaning: Loaded the JSON telemetry file into Tableau. Converted raw timestamp fields into standard datetime dimensions and mapped status messages.
2. Dashboard Construction: Created a primary horizontal bar chart displaying total failure instances aggregated by factory site. Created a secondary split chart breaking down failures by the 9 machine types per location.
3. Diagnostic Insights: Identified Daikibo Factory Meiyo (Tokyo, Japan) as the site with the highest frequency of machine breakdowns. Isolated specific heavy machinery types driving operational downtime in Meiyo.

4.8.2 Task 2 Methodology: Forensic Pay Equity Audit

1. Dataset Review: Ingested `Equality Table.xlsx` consisting of columns: Factory, Job Role, and Equality Score (-100 to +100).
2. Formula Design: Built an advanced nested conditional statement to categorize employee equality scores based on Deloitte's compliance thresholds:
`=IF(AND(C2>=-10, C2<=10), "Fair", IF(OR(AND(C2<-10, C2>=-20), AND(C2>10, C2<=20)), "Unfair", "Highly Discriminative"))`
3. Column Execution & Verification: Created the `Equality Class` column, applied the logic across all records, and validated exact boundary conditions (-20, -10, 0, 10, 20).

4.9 Expected Outcomes & Key Results

The quantitative findings and classification rules established during the simulation are summarized below:

Table 4.1: Classification Threshold Rules for Equality Score

Equality Score Range	Classification Category	Compliance Status	Recommended Action
-10 to +10 (0 is ideal)	Fair	Fully Compliant	Maintain standard salary review cycle
-20 to -11 OR +11 to +20	Unfair	Moderate Risk	Flag for departmental compensation audit
< -20 OR > +20	Highly Discriminative	Critical Violation	Immediate legal & HR forensic investigation

Table 4.2: Comparative Summary of Telemetry Analytics across Client Factories

Factory Location	Country	Breakdown Frequency	Primary Vulnerable Machine Type
Daikibo Factory Meiyo	Tokyo, Japan	Highest Failure Frequency	Laser Cutter Unit / Heavy Assembly Arm
Daikibo Factory Seiko	Osaka, Japan	Moderate Failure Frequency	Automated Stamping Press
Daikibo Berlin	Berlin, Germany	Lowest Failure Frequency	Conveyor Motor Drive
Daikibo Shenzhen	Shenzhen, China	Moderate Failure Frequency	CNC Milling Station

CHAPTER 5: BIBLIOGRAPHY / REFERENCES

1. Deloitte Touche Tohmatsu Limited. (2026). Deloitte Data Analytics and Forensic Technology Simulation Guide. Forage Platform.
2. Tableau Software. (2024). Designing Visual Dashboards for Complex Data Aggregation and IoT Telemetry Analysis. Tableau Knowledge Base.
3. Walkenbach, J. (2021). Microsoft Excel 365 Bible: Advanced Formulas, Functions, and Data Modeling. John Wiley & Sons.
4. Alexander, M., & Kusleika, R. (2022). Excel Data Analysis for Dummies and Business Analysts (5th ed.). Wiley.
5. Few, S. (2019). Show Me the Numbers: Designing Tables and Graphs to Enlighten (2nd ed.). Analytics Press.
6. IILM University School of Computer Science & Engineering. (2026). Guidelines and Evaluation Criteria for B.Tech Internship Reports. Greater Noida, U.P.